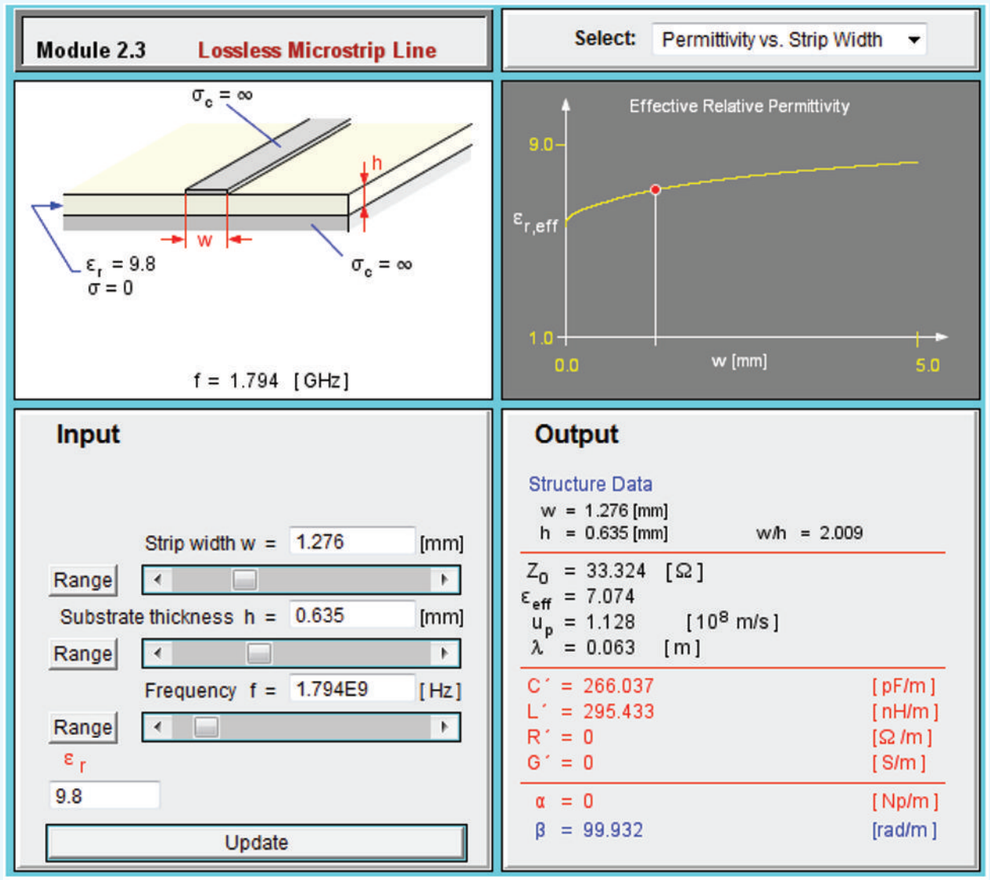
Module 2.3 Lossless Microstrip Line The output panel lists the values of the transmission line parameters and displays the variation of Z_0 and ε_{eff} with h and w.



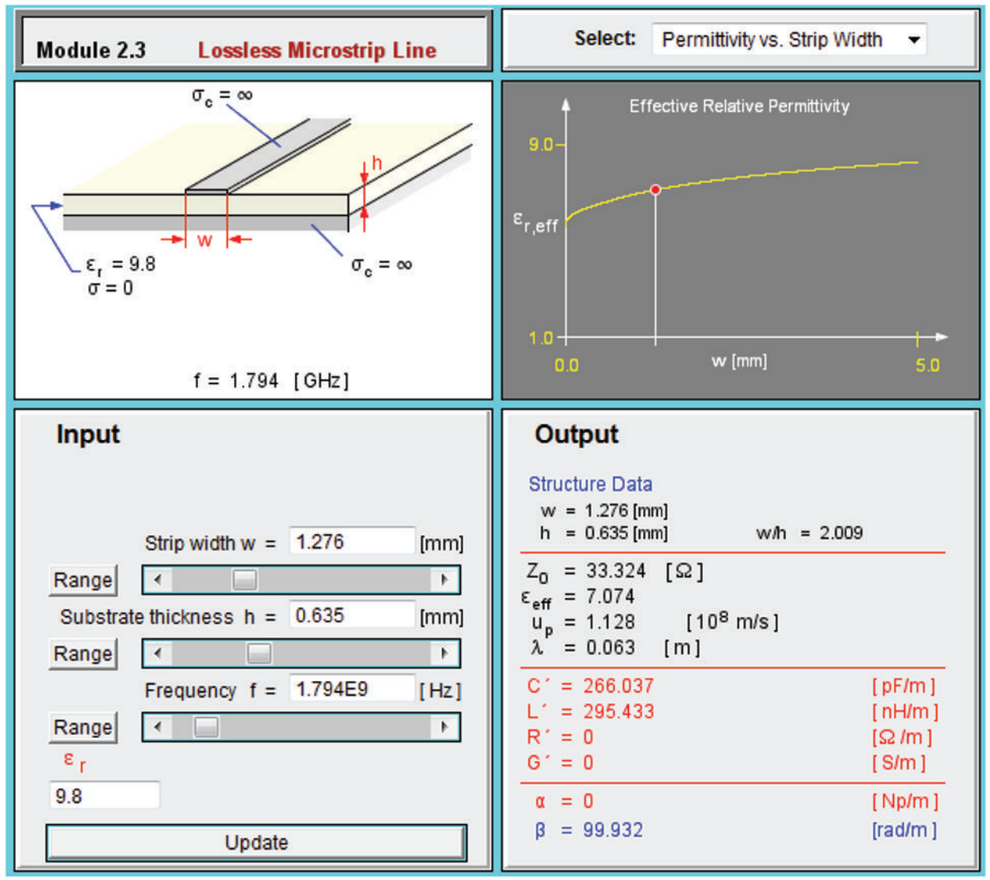
$$\alpha = 0 \qquad \text{(because } R' = G' = 0\text{)}, \tag{2.41e}$$

$$\beta = \frac{\omega}{c}\sqrt{\varepsilon_{\text{eff}}}. \tag{2.41f}$$

2-5.3 Design Process

The preceding expressions allow us to compute the values of Z_0 and the other propagation parameters when given values for ε_r, h, and w. This is exactly what is needed in order to analyze a circuit containing a microstrip transmission line. To perform the reverse process, namely to design a microstrip line by selecting values for its w and h such that their ratio yields the required value of Z_0 (to satisfy design specifications), we need to express s in terms of Z_0. The expression for Z_0 given by Eq. (2.39) is rather complicated, so inverting it to obtain an expression for s in terms of Z_0 is rather difficult. An alternative option is to generate a family of curves similar to those displayed in **Fig. 2-11** and to use them to estimate

模块 2.3 无损微带线 输出面板列出了传输线参数的值，并显示 Z_0 和 ε_{eff} 随 h 和 w 的变化。



$$\alpha = 0 \text{（因为 } R' = G' = 0\text{）}, \tag{2.41e}$$

$$\beta = \frac{\omega}{c}\sqrt{\varepsilon_{\text{eff}}}。 \tag{2.41f}$$

2-5.3 设计流程

当给定 ε_r、h 和 w 的值时，前面的表达式允许我们计算 Z_0 和其他传播参数的值。这正是所需要的分析包含微带传输线的电路。为了执行相反的过程，即通过选择 w 和 h 的值来设计微带线，使得它们的比率产生所需的 Z_0（值以满足设计规范），我们需要用 Z_0 来表达 s。 Z_0 的表达式由等式给出。 (2.39) 相当复杂，因此将其求逆以获得 s 用 Z_0 表示的表达式相当困难。另一种选择是生成类似于图 2-11 中显示的曲线族，并使用它们来估计

s for a specified value of Z_0. A logical extension of the graphical approach is to generate curve-fit expressions that provide high-accuracy estimates of s. The error associated with the following formulas is less than 2%:

(a) For $Z_0 \le (44 - 2\varepsilon_r)\ \Omega$,

$$s = \frac{w}{h} = \frac{2}{\pi}\left\{(q-1) - \ln(2q-1) + \frac{\varepsilon_r - 1}{2\varepsilon_r}\left[\ln(q-1) + 0.29 - \frac{0.52}{\varepsilon_r}\right]\right\} \quad (2.42)$$

with

$$q = \frac{60\pi^2}{Z_0\sqrt{\varepsilon_r}},$$

and

(b) for $Z_0 \ge (44 - 2\varepsilon_r)\ \Omega$,

$$s = \frac{w}{h} = \frac{8e^p}{e^{2p} - 2}, \quad (2.43a)$$

with

$$p = \sqrt{\frac{\varepsilon_r + 1}{2}}\,\frac{Z_0}{60} + \left(\frac{\varepsilon_r - 1}{\varepsilon_r + 1}\right)\left(0.23 + \frac{0.12}{\varepsilon_r}\right). \quad (2.43b)$$

The above expressions presume that ε_r, the relative permittivity of the dielectric substrate, has already been specified. For typical substrate materials including Duroid, Teflon, silicon, and sapphire, ε_r ranges between 2 and 15.

Example 2-2: Microstrip Line

A 50 Ω microstrip line uses a 0.5-mm thick sapphire substrate with $\varepsilon_r = 9$. What is the width of its copper strip?

Solution: Since $Z_0 = 50 > 44 - 18 = 32$, we should use Eq. (2.43):

$$p = \sqrt{\frac{\varepsilon_r + 1}{2}} \times \frac{Z_0}{60} + \left(\frac{\varepsilon_r - 1}{\varepsilon_r + 1}\right)\left(0.23 + \frac{0.12}{\varepsilon_r}\right)$$

$$= \sqrt{\frac{9+1}{2}} \times \frac{50}{60} + \left(\frac{9-1}{9+1}\right)\left(0.23 + \frac{0.12}{9}\right) = 2.06,$$

$$s = \frac{w}{h} = \frac{8e^p}{e^{2p} - 2} = \frac{8e^{2.06}}{e^{4.12} - 2} = 1.056.$$

Hence,

$$w = sh = 1.056 \times 0.5\ \text{mm} = 0.53\ \text{mm}.$$

To check our calculations, we use $s = 1.056$ to calculate Z_0 to verify that the value we obtained is indeed equal or close to 50 Ω. With $\varepsilon_r = 9$, Eqs. (2.37a) to (2.40) yield

$$x = 0.55, \qquad y = 0.99,$$
$$t = 12.51, \qquad \varepsilon_{\text{eff}} = 6.11,$$
$$Z_0 = 49.93\ \Omega.$$

The calculated value of Z_0 is, for all practical purposes, equal to the value specified in the problem statement.

Exercise 2-5: A microstrip transmission line uses a strip of width w and height $h = 1$ mm over a substrate of relative permittivity $\varepsilon_r = 4$. What should w be so that the characteristic impedance of the line is $Z_0 = 50\ \Omega$?

Answer: $w = 2.05$ mm. (See EM.)

2-6 The Lossless Transmission Line: General Considerations

According to the preceding section, a transmission line is fully characterized by two fundamental parameters: its propagation constant γ and its characteristic impedance Z_0, both of which are specified by the angular frequency ω and the line parameters R', L', G', and C'.

► In many practical situations, the transmission line can be designed to exhibit low ohmic losses by selecting conductors with very high conductivities and dielectric materials (separating the conductors) with negligible conductivities. As a result, R' and G' assume very small values such that $R' \ll \omega L'$ and $G' \ll \omega C'$. ◄

These conditions allow us to set $R' = G' \approx 0$ in Eq. (2.22), which yields

$$\gamma = \alpha + j\beta = j\omega\sqrt{L'C'}, \quad (2.44)$$

which in turn implies that

$$\alpha = 0 \qquad \textbf{(lossless line)}, \quad (2.45a)$$
$$\beta = \omega\sqrt{L'C'} \qquad \textbf{(lossless line)}. \quad (2.45b)$$

62 CHAPTER 2 TRANSMISSION LINES

s 为 Z_0 的指定值。图形方法的逻辑扩展是生成提供 s 高精度估计的曲线拟合表达式。下列公式的误差小于2%：

(a) 对于 $Z_0 \le (44 - 2\varepsilon_r)\ \Omega$，

$$s = \frac{w}{h} = \frac{2}{\pi}\left\{(q-1) - \ln(2q-1) + \frac{\varepsilon_r - 1}{2\varepsilon_r}\left[\ln(q-1) + 0.29 - \frac{0.52}{\varepsilon_r}\right]\right\} \quad (2.42)$$

和

$$q = \frac{60\pi^2}{Z_0\sqrt{\varepsilon_r}},$$

和

(b) 对于 $Z_0 \ge (44 - 2\varepsilon_r)\ \Omega$，

$$s = \frac{w}{h} = \frac{8e^p}{e^{2p} - 2}, \quad (2.43a)$$

和

$$p = \sqrt{\frac{\varepsilon_r + 1}{2}}\,\frac{Z_0}{60} + \left(\frac{\varepsilon_r - 1}{\varepsilon_r + 1}\right)\left(0.23 + \frac{0.12}{\varepsilon_r}\right). \quad (2.43b)$$

上述表达式假定介电基板的相对介电常数 ε_r 已被指定。对于包括 Duroid、Teflon、硅和蓝宝石在内的典型基板材料，ε_r 范围在 2 到 15 之间。

例 2-2：微带线

50Ω微带线采用0.5mm厚的蓝宝石基板，$\varepsilon_r = 9$.其铜带宽度是多少?

解：由于$Z_0 = 50 > 44 - 18 = 32$，我们应该使用方程：(2.43)：

$$p = \sqrt{\frac{\varepsilon_r + 1}{2}} \times \frac{Z_0}{60} + \left(\frac{\varepsilon_r - 1}{\varepsilon_r + 1}\right)\left(0.23 + \frac{0.12}{\varepsilon_r}\right)$$

$$= \sqrt{\frac{9+1}{2}} \times \frac{50}{60} + \left(\frac{9-1}{9+1}\right)\left(0.23 + \frac{0.12}{9}\right) = 2.06,$$

$$s = \frac{w}{h} = \frac{8e^p}{e^{2p} - 2} = \frac{8e^{2.06}}{e^{4.12} - 2} = 1.056.$$

因此，

$$w = sh = 1.056 \times 0.5\ \text{mm} = 0.53\ \text{mm}.$$

为了检查我们的计算，我们使用 s = 1.056 来计算 Z_0，以验证我们获得的值确实等于或接近 50 Ω。对于 $\varepsilon_r = 9$，等式。 (2.37a) 至 (2.40) 产率

$$x = 0.55, \qquad y = 0.99,$$
$$t = 12.51, \qquad \varepsilon_{\text{eff}} = 6.11,$$
$$Z_0 = 49.93\ \Omega.$$

出于所有实际目的，Z_0 的计算值等于问题陈述中指定的值。

练习 2-5：微带传输线在相对介电常数为 $\varepsilon_r =$ 的基板上使用宽度为 w、高度为 h = 1 mm 的条带 4. 为了使线路的特性阻抗为 $Z_0 = 50\ \Omega$，w 应该是多少?

答案：w = 2.05 毫米。 （参见EM。）

2-6 无损传输线：一般考虑因素

根据上一节，传输线完全由两个基本参数表征：传播常数 γ 和特性阻抗 Z_0，这两个参数均由角频率 ω 和线路参数 R′、L′、G′ 和 C′ 指定。

► 在许多实际情况下，通过选择电导率非常高的导体和电导率可忽略不计的介电材料（分隔导体），传输线可以设计成具有低欧姆损耗。结果，R′和G′呈现非常小的值，使得$R' \ll \omega L'$和$G' \ll \omega C'$。 ◄

这些条件允许我们在方程（1）中设置 $R' = G' \approx 0$。 （2.22），得出

$$\gamma = \alpha + j\beta = j\omega\sqrt{L'C'}, \quad (2.44)$$

这反过来意味着

$$\alpha = 0 \qquad \textbf{(lossless line)}, \quad (2.45a)$$
$$\beta = \omega\sqrt{L'C'} \qquad \textbf{(lossless line)}. \quad (2.45b)$$

For the characteristic impedance, application of the lossless line conditions to Eq. (2.29) leads to

$$Z_0 = \sqrt{\frac{L'}{C'}} \qquad \textbf{(lossless line)}, \tag{2.46}$$

which now is a real number. Using the lossless line expression for β [Eq. (2.45b)], we obtain the following expressions for the guide wavelength λ and the phase velocity u_p:

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{\omega\sqrt{L'C'}}, \tag{2.47}$$

$$u_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{L'C'}}. \tag{2.48}$$

Upon using Eq. (2.10), Eqs. (2.45b) and (2.48) may be rewritten as

$$\beta = \omega\sqrt{\mu\varepsilon} \qquad \text{(rad/m)}, \tag{2.49}$$

$$u_p = \frac{1}{\sqrt{\mu\varepsilon}} \qquad \text{(m/s)}, \tag{2.50}$$

where μ and ε are, respectively, the magnetic permeability and electrical permittivity of the insulating material separating the conductors. Materials used for this purpose are usually characterized by a permeability $\mu_0 = 4\pi \times 10^{-7}$ H/m (the permeability of free space). Also, the permittivity ε is often specified in terms of the relative permittivity ε_r defined as

$$\varepsilon_r = \varepsilon/\varepsilon_0, \tag{2.51}$$

where $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m $\approx (1/36\pi) \times 10^{-9}$ F/m is the permittivity of free space (vacuum). Hence, Eq. (2.50) becomes

$$u_p = \frac{1}{\sqrt{\mu_0\varepsilon_r\varepsilon_0}} = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \cdot \frac{1}{\sqrt{\varepsilon_r}} = \frac{c}{\sqrt{\varepsilon_r}}, \tag{2.52}$$

where $c = 1/\sqrt{\mu_0\varepsilon_0} = 3 \times 10^8$ m/s is the velocity of light in free space. If the insulating material between the conductors is air, then $\varepsilon_r = 1$ and $u_p = c$. In view of Eq. (2.51) and the relationship between λ and u_p given by Eq. (2.33), the wavelength is given by

$$\lambda = \frac{u_p}{f} = \frac{c}{f}\frac{1}{\sqrt{\varepsilon_r}} = \frac{\lambda_0}{\sqrt{\varepsilon_r}}, \tag{2.53}$$

where $\lambda_0 = c/f$ is the wavelength in air corresponding to a frequency f. Note that, because both u_p and λ depend on ε_r, the choice of the type of insulating material used in a transmission line is dictated not only by its mechanical properties but by its electrical properties as well.

According to Eq. (2.52), if ε_r of the insulating material is independent of f (which usually is the case for commonly used TEM lines), the same independence applies to u_p.

► If sinusoidal waves of different frequencies travel on a transmission line with the same phase velocity, the line is called ***nondispersive***. ◄

This is an important feature to consider when digital data are transmitted in the form of pulses. A rectangular pulse or a series of pulses is composed of many Fourier components with different frequencies. If the phase velocity is the same for all frequency components (or at least for the dominant ones), then the pulse's shape does not change as it travels down the line. In contrast, the shape of a pulse propagating in a dispersive medium becomes progressively distorted, and the pulse length increases (stretches out) as a function of the distance traveled in the medium (**Fig. 2-3**), thereby imposing a limitation on the maximum data rate (which is related to the length of the individual pulses and the spacing between adjacent pulses) that can be transmitted through the medium without loss of information.

Table 2-2 provides a list of the expressions for γ, Z_0, and u_p for the general case of a lossy line and for several types of lossless lines. The expressions for the lossless lines are based on the equations for L' and C' given in **Table 2-1**.

Exercise 2-6: For a lossless transmission line, $\lambda = 20.7$ cm at 1 GHz. Find ε_r of the insulating material.

Answer: $\varepsilon_r = 2.1$. (See EM.)

Exercise 2-7: A lossless transmission line uses a dielectric insulating material with $\varepsilon_r = 4$. If its line capacitance is $C' = 10$ (pF/m), find (a) the phase velocity u_p, (b) the line inductance L', and (c) the characteristic impedance Z_0.

Answer: (a) $u_p = 1.5 \times 10^8$ (m/s), (b) $L' = 4.45$ (μH/m), (c) $Z_0 = 667.1\ \Omega$. (See EM.)

2-6.1 Voltage Reflection Coefficient

With $\gamma = j\beta$ for the lossless line, Eqs. (2.26a) and (2.30) for the total voltage and current become

$$\widetilde{V}(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z}, \tag{2.54a}$$

对于特性阻抗，将无损线路条件应用到等式： (2.29) 导致

$$Z_0 = \sqrt{\frac{L'}{C'}} \qquad \textbf{(lossless line)}, \tag{2.46}$$

现在是一个实数。使用 β 的无损线表达式 [Eq. (2.45b)]，我们得到波导波长λ 和相速度u_p的表达式：

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{\omega\sqrt{L'C'}}, \tag{2.47}$$

$$u_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{L'C'}}. \tag{2.48}$$

使用方程式后。 (2.10) ，等式。 (2.45b) 和 (2.48) 可以重写为

$$\beta = \omega\sqrt{\mu\varepsilon} \qquad \text{(rad/m)}, \tag{2.49}$$

$$u_p = \frac{1}{\sqrt{\mu\varepsilon}} \qquad \text{(m/s)}, \tag{2.50}$$

其中 μ 和 ε 分别是分隔导体的绝缘材料的磁导率和介电常数。用于此目的的材料通常以磁导率 $\mu_0 = 4\pi \times 10^{-7}$ H/m（自由空间的磁导率）为特征。此外，介电常数 ε 通常用相对介电常数 ε_r 来指定，定义为

$$\varepsilon_r = \varepsilon/\varepsilon_0, \tag{2.51}$$

其中 $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m $\approx (1/36\pi) \times 10^{-9}$ F/m 是自由空间（真空）的介电常数。因此，等式。 (2.50) 变为

$$u_p = \frac{1}{\sqrt{\mu_0\varepsilon_r\varepsilon_0}} = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \cdot \frac{1}{\sqrt{\varepsilon_r}} = \frac{c}{\sqrt{\varepsilon_r}}, \tag{2.52}$$

其中 $c = 1/\sqrt{\mu_0\varepsilon_0} = 3 \times 10^8$ m/s 是自由空间中的光速。如果导体之间的绝缘材料是空气，则$\varepsilon_r = 1$ 和$u_p = c$。鉴于方程。 (2.51) 以及方程 (2.51) 给出的 λ 和 u_p 之间的关系(2.33)，波长由下式给出

$$\lambda = \frac{u_p}{f} = \frac{c}{f}\frac{1}{\sqrt{\varepsilon_r}} = \frac{\lambda_0}{\sqrt{\varepsilon_r}}, \tag{2.53}$$

其中 $\lambda_0 = c/f$ 是空气中对应于频率 f 的波长。请注意，由于 u_p 和 λ 都依赖于 ε_r，因此

传输线中使用的绝缘材料类型的选择不仅取决于其机械性能，还取决于其电气性能。

根据方程。根据方程 (2.52)，如果绝缘材料的 ε_r 与 f 无关（对于常用的 TEM 线通常就是这种情况），则同样的独立性也适用于 u_p。

► 如果不同频率的正弦波以相同的相速度在传输线上传播，则该传输线称为非色散传输线。 ◄

当数字数据以脉冲形式传输时，这是一个需要考虑的重要特征。矩形脉冲或一系列脉冲由许多不同频率的傅立叶分量组成。如果所有频率分量（或至少主要频率分量）的相速度相同，则脉冲的形状在沿线路传播时不会改变。相比之下，在色散介质中传播的脉冲形状逐渐扭曲，并且脉冲长度随着在介质中传播的距离而增加（拉长）（图2-3），从而对可以通过介质传输而不丢失信息的最大数据速率（与单个脉冲的长度和相邻脉冲之间的间距有关）施加限制。

表 2-2 提供了有损线路的一般情况和几种类型的无损线路的 γ、Z_0 和 u_p 的表达式列表。无损线路的表达式基于表 2-1 中给出的 L′ 和 C′ 方程。

练习 2-6：对于无损传输线，1 GHz 时 λ = 20.7 cm。找到绝缘材料的ε_r。

Answer: $\varepsilon_r = 2.1$. (See EM.)

练习 2-7：无损传输线使用 ε_r = 的电介质绝缘材料 4。如果其线路电容为 C′ = 10 (pF/m)，则求 (a) 相速度 u_p，(b) 线路电感 L′，以及 (c) 特性阻抗 Z_0。

答案：(a) $u_p = 1.5 \times 10^8$ (m/s)，(b) L′ = 4.45 (μ H/m)，(c) $Z_0 = 667.1\ \Omega$。 (参见EM。)

2-6.1 电压反射系数

对于无损线，方程为 γ = jβ。 (2.26a) 和 (2.30) 的总电压和电流变为

$$\widetilde{V}(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z}, \tag{2.54a}$$

Table 2-2 Characteristic parameters of transmission lines.

	Propagation Constant $\gamma = \alpha + j\beta$	Phase Velocity u_p	Characteristic Impedance Z_0
General case	$\gamma = \sqrt{(R' + j\omega L')(G' + j\omega C')}$	$u_p = \omega/\beta$	$Z_0 = \sqrt{\dfrac{(R' + j\omega L')}{(G' + j\omega C')}}$
Lossless $(R' = G' = 0)$	$\alpha = 0,\ \beta = \omega\sqrt{\varepsilon_r}/c$	$u_p = c/\sqrt{\varepsilon_r}$	$Z_0 = \sqrt{L'/C'}$
Lossless coaxial	$\alpha = 0,\ \beta = \omega\sqrt{\varepsilon_r}/c$	$u_p = c/\sqrt{\varepsilon_r}$	$Z_0 = (60/\sqrt{\varepsilon_r})\ln(b/a)$
Lossless two-wire	$\alpha = 0,\ \beta = \omega\sqrt{\varepsilon_r}/c$	$u_p = c/\sqrt{\varepsilon_r}$	$Z_0 = (120/\sqrt{\varepsilon_r}) \cdot \ln\left[(D/d) + \sqrt{(D/d)^2 - 1}\right]$ $Z_0 \approx (120/\sqrt{\varepsilon_r})\ln(2D/d)$, if $D \gg d$
Lossless parallel-plate	$\alpha = 0,\ \beta = \omega\sqrt{\varepsilon_r}/c$	$u_p = c/\sqrt{\varepsilon_r}$	$Z_0 = (120\pi/\sqrt{\varepsilon_r})\,(h/w)$

Notes: (1) $\mu = \mu_0,\ \varepsilon = \varepsilon_r\varepsilon_0,\ c = 1/\sqrt{\mu_0\varepsilon_0}$, and $\sqrt{\mu_0/\varepsilon_0} \approx (120\pi)\ \Omega$, where ε_r is the relative permittivity of insulating material. (2) For coaxial line, a and b are radii of inner and outer conductors. (3) For two-wire line, d = wire diameter and D = separation between wire centers. (4) For parallel-plate line, w = width of plate and h = separation between the plates.

$$\tilde{I}(z) = \frac{V_0^+}{Z_0}e^{-j\beta z} - \frac{V_0^-}{Z_0}e^{j\beta z}. \tag{2.54b}$$

These expressions contain two unknowns, V_0^+ and V_0^-. According to Section 1-7.2, an exponential factor of the form $e^{-j\beta z}$ is associated with a wave traveling in the positive z direction, from the source (sending end) to the load (receiving end). Accordingly, we refer to it as the ***incident wave*** with V_0^+ as its voltage amplitude. Similarly, the term containing $V_0^- e^{j\beta z}$ represents a ***reflected wave*** with voltage amplitude V_0^- traveling along the negative z direction from the load to the source.

To determine V_0^+ and V_0^-, we need to consider the lossless transmission line in the context of the complete circuit, including a generator circuit at its input terminals and a load at its output terminals, as shown in **Fig. 2-12**. The line of length l is terminated in an arbitrary ***load impedance*** Z_L.

► For mathematical convenience, the reference of the spatial coordinate z is chosen such that $z = 0$ corresponds to the location of the load, not the generator. ◄

At the sending end, at $z = -l$, the line is connected to a sinusoidal voltage source with phasor voltage $\widetilde{V}_g$ and internal impedance Z_g. Since z points from the generator to the load, positive values of z correspond to locations beyond the load; therefore, they are irrelevant to our circuit. In future sections, we will find it more convenient to work with a spatial dimension that also starts at the load but whose direction is opposite of z. We shall call it the ***distance from the load*** d and define it as $d = -z$, as shown in **Fig. 2-12**.

The phasor voltage across the load, $\widetilde{V}_L$, and the phasor current through it, $\tilde{I}_L$, are related by the load impedance Z_L as

$$Z_L = \frac{\widetilde{V}_L}{\tilde{I}_L}. \tag{2.55}$$

The voltage $\widetilde{V}_L$ is the total voltage on the line $\widetilde{V}(z)$ given by Eq. (2.54a), and $\tilde{I}_L$ is the total current $\tilde{I}(z)$ given by Eq. (2.54b). Both are evaluated at $z = 0$:

$$\widetilde{V}_L = \widetilde{V}(z{=}0) = V_0^+ + V_0^-, \tag{2.56a}$$

表2-2 传输线特性参数。

	Propagation Constant $\gamma = \alpha + j\beta$	Phase Velocity u_p	Characteristic Impedance Z_0
General case	$\gamma = \sqrt{(R' + j\omega L')(G' + j\omega C')}$	$u_p = \omega/\beta$	$Z_0 = \sqrt{\dfrac{(R' + j\omega L')}{(G' + j\omega C')}}$
Lossless $(R' = G' = 0)$	$\alpha = 0,\ \beta = \omega\sqrt{\varepsilon_r}/c$	$u_p = c/\sqrt{\varepsilon_r}$	$Z_0 = \sqrt{L'/C'}$
Lossless coaxial	$\alpha = 0,\ \beta = \omega\sqrt{\varepsilon_r}/c$	$u_p = c/\sqrt{\varepsilon_r}$	$Z_0 = (60/\sqrt{\varepsilon_r})\ln(b/a)$
Lossless two-wire	$\alpha = 0,\ \beta = \omega\sqrt{\varepsilon_r}/c$	$u_p = c/\sqrt{\varepsilon_r}$	$Z_0 = (120/\sqrt{\varepsilon_r}) \cdot \ln\left[(D/d) + \sqrt{(D/d)^2 - 1}\right]$ $Z_0 \approx (120/\sqrt{\varepsilon_r})\ln(2D/d)$, if $D \gg d$
Lossless parallel-plate	$\alpha = 0,\ \beta = \omega\sqrt{\varepsilon_r}/c$	$u_p = c/\sqrt{\varepsilon_r}$	$Z_0 = (120\pi/\sqrt{\varepsilon_r})\,(h/w)$

Notes: (1) $\mu = \mu_0,\ \varepsilon = \varepsilon_r\varepsilon_0,\ c = 1/\sqrt{\mu_0\varepsilon_0}$, and $\sqrt{\mu_0/\varepsilon_0} \approx (120\pi)\ \Omega$, where ε_r is the relative permittivity of insulating material. (2) For coaxial line, a and b are radii of inner and outer conductors. (3) For two-wire line, d = wire diameter and D = separation between wire centers. (4) For parallel-plate line, w = width of plate and h = separation between the plates.

$$\tilde{I}(z) = \frac{V_0^+}{Z_0}e^{-j\beta z} - \frac{V_0^-}{Z_0}e^{j\beta z}. \tag{2.54b}$$

这些表达式包含两个未知数 V_0^+ 和 V_0^-。根据第 1-7.2 节，$e^{-j\beta z}$ 形式的指数因子与沿正 z 方向从源（发送端）到负载（接收端）传播的波相关。因此，我们将其称为以 V_0^+ 为其电压幅度的入射波。类似地，包含 $V_0^- e^{j\beta z}$ 的项表示电压幅度为 V_0^- 的反射波，沿负 z 方向从负载传播到源。

为了确定 V_0^+ 和 V_0^-，我们需要考虑完整电路中的无损传输线，包括输入端的发生器电路和输出端的负载，如图 2-12 所示。长度为 l 的线路以任意负载阻抗 Z_L 终止。

► 为了数学方便，选择空间坐标 z 的参考，使得 z = 0 对应于负载的位置，而不是发电机的位置。◄

在发送端，在 $z = -l$ 处，线路连接到具有相量电压 $\widetilde{V}_g$ 和内部阻抗 Z_g 的正弦电压源。由于 z 点从发电机到负载，因此 z 的正值对应于超出负载的位置；因此，它们与我们的电路无关。在以后的章节中，我们会发现使用同样从载荷开始但方向与 z 相反的空间维度会更方便。我们将其称为距负载的距离 d，并将其定义为 $d = -z$，如图 2-12 所示。

负载两端的相量电压 $\widetilde{V}_L$ 和流过负载的相量电流 $\tilde{I}_L$ 与负载阻抗 Z_L 的关系如下：

$$Z_L = \frac{\widetilde{V}_L}{\tilde{I}_L}. \tag{2.55}$$

电压 $\widetilde{V}_L$ 是等式 1 给出的线上 $\widetilde{V}(z)$ 上的总电压。(2.54a)，$\tilde{I}_L$ 是由式 (2.54a) 给出的总电流 $\tilde{I}(z)$。(2.54b)。两者均在 $z = 0$ 处求值：

$$\widetilde{V}_L = \widetilde{V}(z{=}0) = V_0^+ + V_0^-, \tag{2.56a}$$

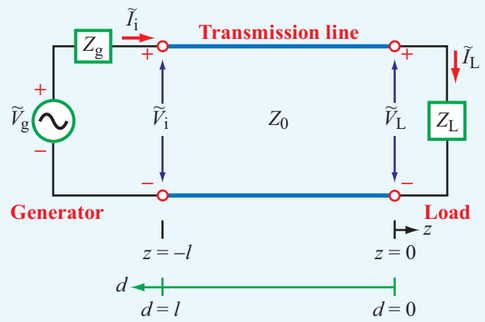


Figure 2-12 Transmission line of length l connected on one end to a generator circuit and on the other end to a load Z_L. The load is located at $z = 0$, and the generator terminals are at $z = -l$. Coordinate d is defined as $d = -z$.

$$\tilde{I}_L = \tilde{I}(z{=}0) = \frac{V_0^+}{Z_0} - \frac{V_0^-}{Z_0}. \tag{2.56b}$$

Using these expressions in Eq. (2.55), we obtain

$$Z_L = \left(\frac{V_0^+ + V_0^-}{V_0^+ - V_0^-}\right) Z_0. \tag{2.57}$$

Solving for V_0^- gives

$$V_0^- = \left(\frac{Z_L - Z_0}{Z_L + Z_0}\right) V_0^+. \tag{2.58}$$

► The ratio of the amplitudes of the reflected and incident voltage waves at the load is known as the ***voltage reflection coefficient*** Γ. ◄

From Eq. (2.58), it follows that

$$\Gamma = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{Z_L/Z_0 - 1}{Z_L/Z_0 + 1} = \frac{z_L - 1}{z_L + 1} \quad \textbf{(dimensionless)}, \tag{2.59}$$

where

$$z_L = \frac{Z_L}{Z_0} \tag{2.60}$$

is the ***normalized load impedance***. In many transmission line problems, we can streamline the necessary computation by normalizing all impedances in the circuit to the characteristic impedance Z_0. *Normalized impedances are denoted by lowercase letters.*

In view of Eq. (2.28), the ratio of the current amplitudes is

$$\frac{I_0^-}{I_0^+} = -\frac{V_0^-}{V_0^+} = -\Gamma. \tag{2.61}$$

► We note that whereas the ratio of the voltage amplitudes is equal to Γ, the ratio of the current amplitudes is equal to $-\Gamma$. ◄

The reflection coefficient Γ is governed by a single parameter, the normalized load impedance z_L. As indicated by Eq. (2.46), Z_0 of a lossless line is a real number. However, Z_L is in general a complex quantity, as in the case of a series RL circuit, for example, for which $Z_L = R + j\omega L$. Hence, in general Γ also is complex and given by

$$\Gamma = |\Gamma| e^{j\theta_r}, \tag{2.62}$$

where $|\Gamma|$ is the magnitude of Γ and θ_r is its phase angle. Note that $|\Gamma| \leq 1$.

► A load is said to be ***matched*** to a transmission line if $Z_L = Z_0$ because then there will be no reflection by the load ($\Gamma = 0$ and $V_0^- = 0$). ◄

On the other hand, when the load is an open circuit ($Z_L = \infty$), $\Gamma = 1$ and $V_0^- = V_0^+$, and when it is a short circuit ($Z_L = 0$), $\Gamma = -1$ and $V_0^- = -V_0^+$ (**Table 2-3**).

Example 2-3: Reflection Coefficient of a Series RC Load

A 100 Ω transmission line is connected to a load consisting of a 50 Ω resistor in series with a 10 pF capacitor. Find the reflection coefficient at the load for a 100 MHz signal.

Solution: The following quantities are given (**Fig. 2-13**):

$$R_L = 50\ \Omega, \qquad C_L = 10\ \text{pF} = 10^{-11}\ \text{F},$$

$$Z_0 = 100\ \Omega, \qquad f = 100\ \text{MHz} = 10^8\ \text{Hz}.$$

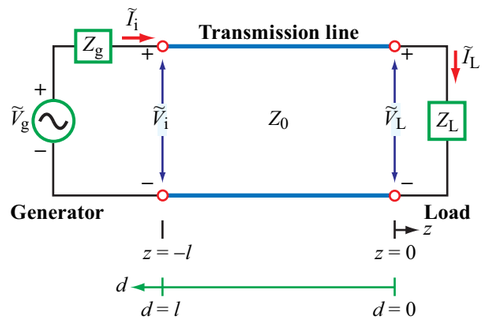


图 2-12 长度为 l 的传输线，一端连接到发电机电路，另一端连接到负载 Z_L。负载位于z=0，发电机端子位于z= –l。坐标 d 定义为 d = –z。

$$\tilde{I}_L = \tilde{I}(z{=}0) = \frac{V_0^+}{Z_0} - \frac{V_0^-}{Z_0}. \tag{2.56b}$$

在等式中使用这些表达式。 (2.55)，我们得到

$$Z_L = \left(\frac{V_0^+ + V_0^-}{V_0^+ - V_0^-}\right) Z_0. \tag{2.57}$$

求解 V_0^- 给出

$$V_0^- = \left(\frac{Z_L - Z_0}{Z_L + Z_0}\right) V_0^+. \tag{2.58}$$

► 负载处反射电压波和入射电压波的振幅之比称为电压反射系数Γ。 ◄

从方程。 (2.58)，由此可知

$$\Gamma = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{Z_L/Z_0 - 1}{Z_L/Z_0 + 1} = \frac{z_L - 1}{z_L + 1} \quad \textbf{(dimensionless)}, \tag{2.59}$$

在哪里

$$z_L = \frac{Z_L}{Z_0} \tag{2.60}$$

是归一化负载阻抗。在许多传输线问题中，我们可以通过将电路中的所有阻抗归一化为特性阻抗 Z_0 来简化必要的计算。归一化阻抗用小写字母表示。

鉴于方程。 (2.28) 中，电流幅值之比为

$$\frac{I_0^-}{I_0^+} = -\frac{V_0^-}{V_0^+} = -\Gamma. \tag{2.61}$$

►我们注意到，电压幅值之比等于Γ，而电流幅值之比等于−Γ。 ◄

反射系数 Γ 由单个参数（归一化负载阻抗 z_L）控制。如方程所示。 (2.46)，无损线的Z_0是实数。然而，Z_L 通常是复数，例如在串联 RL 电路的情况下，其中 $Z_L = R + j\omega L$。因此，一般来说 Γ 也是复数，由下式给出

$$\Gamma = |\Gamma| e^{j\theta_r}, \tag{2.62}$$

其中 $|\Gamma|$ 是 Γ 的大小，θ_r 是其相位角。请注意 $|\Gamma| \leq 1$。

► 如果 $Z_L = Z_0$，则称负载与传输线匹配，因为负载不会产生反射（$\Gamma = 0$ 和 $V_0^- = 0$）。 ◄

另一方面，当负载开路（$Z_L = \infty$）时，$\Gamma = 1$和$V_0^- = V_0^+$，当负载短路（$Z_L = 0$）时，$\Gamma = -1$和$V_0^- = -V_0^+$ (表2-3)。

示例 2-3：串联 RC 负载的反射系数

100 Ω 传输线连接到由 50 Ω 电阻器与 10 pF 电容器串联组成的负载。求 100 MHz 信号负载处的反射系数。

解：给出下列数量（图2-13）：

$$R_L = 50\ \Omega, \qquad C_L = 10\ \text{pF} = 10^{-11}\ \text{F},$$

$$Z_0 = 100\ \Omega, \qquad f = 100\ \text{MHz} = 10^8\ \text{Hz}.$$